

Mohammad Choupan Results:

Model: SAG-POOL

Model Net10 :

Train: 93%

Validation : 7% ~260

Test: Test Folder

Random seed : 42 - numpy,Torch

Epochs: 200

Model Net40:

Train: 97%

Validation : 3%

Test: Test Folder

Random seed : 42 - numpy,Torch

Epochs: 200

Tuning :

Features: (x,y,z)

Dataset	Learning rate	Weight decay	Accuracy%
ModelNet10	0.01	0.0005	94.27
ModelNet10	0.02	0.0005	94.27
ModelNet10	0.03	0.0005	93.55
ModelNet10	0.01	0.0003	94.27
ModelNet10	0.01	0.0008	93.55
ModelNet10	0.025	0.0005	93.55
ModelNet10	0.01	0.0002	91.76
ModelNet40	0.02	0.0005	79.54
ModelNet40	0.005	0.0005	82.87
ModelNet40	0.01	0.0005	82.15
ModelNet40	0.01	0.0003	82.15
ModelNet40	0.01	0.0008	79.83
ModelNet40	0.005	0.0003	79.83
ModelNet40	0.004	0.0005	83.14

Final Result:

Dataset	Feature	Learning rate	Weight decay	Accuracy%
ModelNet10	x,y,z	0.01	0.005	87.11
ModelNet10	Between	0.01	0.005	86.12
ModelNet10	Closeness	0.01	0.005	83.26
ModelNet10	Eigen Vector	0.01	0.005	87.33
ModelNet10	Harmonic	0.01	0.005	87.89
ModelNet10	Katz	0.01	0.005	83.04
ModelNet10	Load	0.01	0.005	87.56
ModelNet10	Page Rank	0.01	0.005	87.22
ModelNet10	all	0.01	0.005	87.78
ModelNet40	x,y,z	0.004	0.005	78.44
ModelNet40	Between	0.004	0.005	74.11
ModelNet40	Closeness	0.004	0.005	74.8
ModelNet40	Eigen Vector	0.004	0.005	68.48
ModelNet40	Harmonic	0.004	0.005	77.03
ModelNet40	Katz	0.004	0.005	71.47
ModelNet40	Load	0.004	0.005	71.8
ModelNet40	Page Rank	0.004	0.005	71.27
ModelNet40	all	0.004	0.005	75.04

Mohadese Atyabi Results:

Model : Point-Net

Tuning:

Dataset	Learning rate	Weight decay	Accuracy
ModelNet10	0.1	0.00001	92.18
ModelNet10	0.2	0.00001	91.19
ModelNet10	0.01	0.00001	92.18
ModelNet10	0.02	0.0001	92.62
ModelNet10	0.02	0.001	91.85
ModelNet10	0.02	0.00001	92.95
ModelNet40	0.1	0.00001	83.71
ModelNet40	0.2	0.00001	77.31
ModelNet40	0.02	0.0001	87.44
ModelNet40	0.02	0.001	86.67
ModelNet40	0.02	0.00001	86.35

Final Result:

Dataset	Feature	Learning rate	Weight decay	Accuracy
ModelNet10	Betweenness	0.02	0.00001	92.29
ModelNet10	Closeness centrality	0.02	0.00001	92.73
ModelNet10	Eigenvector centrality	0.02	0.00001	92.51
ModelNet10	Harmonic centrality	0.02	0.00001	92.29
ModelNet10	Katz centrality	0.02	0.00001	92.62
ModelNet10	Load centrality	0.02	0.00001	92.29
ModelNet10	Pagerank	0.02	0.00001	92.4

Model : Feature Concat (100epoch)

ModelNet10	xyz	0.01	0.0005	94.34
ModelNet10	all	0.01	0.0005	92.29
ModelNet40	XYZ	0.01	0.0005	87.6
ModelNet40	all	0.01	0.0005	86.67

Model : Point Net based Graph Polling(100epoch)

ModelNet10	xyz	0.01	0.0005	92.23
ModelNet10	all	0.01	0.0005	95.8
ModelNet40	XYZ	0.01	0.0005	87.6
ModelNet40	all	0.01	0.0005	84.07

New result : Feature Concat(100 epoch)

ModelNet10	harmonic	0.01	0.0005	92.26
ModelNet40	harmonic	0.01	0.0005	88.2

New result :Point Net based Graph Polling(100epoch)

ModelNet10	harmonic	0.01	0.0005	93.74
ModelNet40	harmonic	0.01	0.0005	85.47